



Data Loading & Visualization Notes (Pandas + Seaborn + Matplotlib)



Load Data

CSV File

```
import pandas as pd

df = pd.read_csv("data.csv")
```

Excel File

```
df = pd.read_excel("data.xlsx")
```

JSON File

```
df = pd.read_json("data.json")
```

SQL Database

```
import sqlite3

conn = sqlite3.connect("database.db")
df = pd.read_sql("SELECT * FROM students", conn)
```

Built-in Seaborn Dataset

```
import seaborn as sns

tips = sns.load_dataset("tips")
iris = sns.load_dataset("iris")
titanic = sns.load_dataset("titanic")
penguins = sns.load_dataset("penguins")
diamonds = sns.load_dataset("diamonds")
```



Data Visualization Plots

1. Scatter Plot

```
sns.scatterplot(data=df, x="age", y="salary")
```

Purpose: Shows the relationship between two numerical variables.

2. Line Plot

```
sns.lineplot(data=df, x="month", y="sales")
```

Purpose: Displays trends or changes over time.

3. Bar Plot

```
sns.barplot(data=df, x="department", y="salary")
```

Purpose: Compares average values across categories.

4. Count Plot

```
sns.countplot(data=df, x="gender")
```

Purpose: Counts the number of records in each category.

5. Histogram

```
sns.histplot(data=df, x="age", kde=True)
```

Purpose: Shows the distribution of a numerical column.

6. Box Plot

```
sns.boxplot(data=df, x="gender", y="salary")
```

Purpose: Displays data spread, median, quartiles, and outliers.

7. Violin Plot

```
sns.violinplot(data=df, x="gender", y="salary")
```

Purpose: Shows both the distribution and density of data.

8. Pie Chart

```
plt.pie(df["gender"].value_counts(), labels=df["gender"].value_counts().index,  
autopct="%1.1f%%")  
plt.show()
```

Purpose: Displays the percentage of each category.

9. Heatmap

```
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="coolwarm")
```

Purpose: Visualizes the correlation between numerical columns.

10. Pair Plot

```
sns.pairplot(df)
```

Purpose: Shows pairwise relationships among numerical features.

11. KDE Plot

```
sns.kdeplot(data=df, x="age", fill=True)
```

Purpose: Displays a smooth probability density curve.

12. Joint Plot

```
sns.jointplot(data=df, x="age", y="salary", kind="scatter")
```

Purpose: Combines a scatter plot with the distributions of both variables.

13. Strip Plot

```
sns.stripplot(data=df, x="gender", y="salary")
```

Purpose: Shows the distribution of individual data points.

14. Swarm Plot

```
sns.swarmplot(data=df, x="gender", y="salary")
```

Purpose: Displays all data points without overlapping.

15. Regression Plot

```
sns.regplot(data=df, x="age", y="salary")
```

Purpose: Shows the relationship between two variables with a regression trend line.

Summary

Plot	Use
Scatter Plot	Relationship between two numerical variables
Line Plot	Trends over time
Bar Plot	Compare categories
Count Plot	Count categories
Histogram	Distribution of data
Box Plot	Spread and outliers
Violin Plot	Distribution + density
Pie Chart	Percentage composition
Heatmap	Correlation analysis
Pair Plot	Relationships between all numeric features
KDE Plot	Probability density
Joint Plot	Scatter + distributions
Strip Plot	Individual observations
Swarm Plot	Non-overlapping observations
Regression Plot	Trend line and relationship